Lecture 3 (General and Standard Forms and LP Formulation)

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General Form

Earlier in the class, I gave a definition of an LP. This wasn't the most general definition or the way computers typically represent LPs. It is equivalent however. Today, we will show the most general form and show the equivalence to standard form. The **General Form** of an LP is the following:

$$\text{minimize (maximize) } z = \vec{c}^T\vec{x}$$

subject to the following **constraints**:

$$\vec{a}_i^T\vec{x} \geq b_i,\ i \in M_1$$

$$\vec{a}_i^T\vec{x} \leq b_i,\ i \in M_2$$

$$\vec{a}_i^T\vec{x} = b_i,\ i \in M_3$$

and the following **sign constraints**:

$$x_j \geq 0,\ j \in N_1$$

$$x_j \leq 0,\ j \in N_2$$

$$x_j \in \mathbb{R},\ j \in N_3$$

A few notes about this formulation. First, we should be clear. x_j is the jth entry in vector $\vec{x}$. Each $\vec{a}_i$ is a separate vector. We also require that each M_i be disjoint from every other M_i. M_1 is the set of all indices where constraints have a $\geq$. M_2 is for $\leq$ constraints. M_3 for equality constraints. As each constraint is only one of the 3 types, the corresponding index will lie in exactly one of the three sets. Often the objective function will replace z with $f(\vec{x})$. A similar think occurs with N_1, N_2 and N_3. As one final thought, we can convert between a minimization and a maximization problem by noting

$$\min -z = \max z$$

We call z the **objective** or **objective function**. Another way that the general form is stated is in terms of matrices and matrix inequalities. Consider the equivalent statement:

$$\text{minimize } z = \vec{c}^T\vec{x}$$

subject to the following **constraints**:

$$A_1\vec{x} \geq \vec{b}_1$$

$$A_2\vec{x} \leq \vec{b}_2$$

$$A_3\vec{x} = \vec{b}_3$$

and the following **sign constraints**:

$$x_j \geq 0,\ j \in N_1$$

$$x_j \leq 0,\ j \in N_2$$

$$x_j \in \mathbb{R},\ j \in N_3$$

Here, A_1, A_2, A_3 are matrices that replace each of the $\vec{a}_i$ and the $\vec{b}$ vectors absorb the corresponding b_i terms.

Standard Form

Usually, we will try to write LPs in standard form. Standard form is used when using the simplex algorithm. **Standard Form** is stated below:

$$\text{minimize } z = \vec{c}^T \vec{x}$$

subject to the following **constraint**:

$$A\vec{x} = \vec{b}$$

and the following **sign constraint**:

$$\vec{x} \geq 0$$

As a reminder, $\vec{x} \geq 0$ means each entry of $\vec{x}$ is greater than or equal to 0. To avoid confusion, this is often written as $\vec{x} \geq \vec{0}$. If the arrow on top of the zero isn't present, this is assumed to be element-wise.

Equivalence of Forms

Now, we will show that general form is equivalent to standard form. To do so, we will transform each of the constraints and sign constraints.

1) Consider constraints of the form

$$\vec{a}_i^T \vec{x} \geq b_i, \ i \in M_1$$

Then, we can subtract a **Slack Variable** to each constraint of this form.

$$\vec{a}_i^T \vec{x} - s_i = b_i, \ i \in M_1, \ s_i \geq 0$$

This slack variable captures how much excess is present on the left side of the equation.

2) Consider constraints of the form

$$\vec{a}_i^T \vec{x} \leq b_i, \ i \in M_2$$

This time, we add a slack variable to the constraint:

$$\vec{a}_i^T \vec{x} + s_i = b_i, \ i \in M_2$$

This time, the slack variable captures how much deficit is on the left side of the equation.

3) Consider sign constraints of the form

$$x_j \leq 0, \ j \in N_2$$

We can flip the inequality by multiplying by -1 on both sides.

$$-x_j \geq 0, \ j \in N_2$$

Then, we use $-x_j$ in place of x_j in each of the constraints.

4) Consider sign constraints of the form

$$x_j \in \mathbb{R}, \ j \in N_3$$

We can replace each x_j in the constraints and sign constraints with two variables by taking

$$x_j = x_j^+ - x_j^-, \ x_j^+, x_j^- \geq 0$$

In theory there are infinitely many values that can be taken by x_j^+ and x_j^-. In practice, row reduction will force only one of these variables to be non-zero. We will see this later when we start solving these LPs. Applying **1)-4)** will turn any general form LP into standard form.

Examples of LPs

Now, let's model a couple of problems as LPs. We will model the LP, then convert to standard form for practice.

Ex 1)

Farmer Bob needs to figure out which crops to grow on his farm. He also has limitations on how many seeds of each variety he can grow. The profit (in thousands of dollars per acre) and the amount of seeds available (each unit of seeds will cover one acre) is given in the table below:

Crop	Beans	Corn	Potato
Decision Variable (Seed Quantity Purchased)	x_1	x_2	x_3
Profit	\$10	\$5	\$12
Quantity	80	100	60

Assume his farm is 200 acres. Formulate this problem as an LP.

General Form Formulation

$$\text{maximize } z = 10x_1 + 5x_2 + 12x_3$$

subject to

$$x_1 \leq 80$$

$$x_2 \leq 100$$

$$x_3 \leq 60$$

and

$$x_1, x_2, x_3 \geq 0$$

Standard Form Formulation

Now we will convert this to standard form:

$$\text{minimize } z = -10x_1 - 5x_2 - 12x_3$$

subject to

$$x_1 + s_1 = 80$$

$$x_2 + s_2 = 100$$

$$x_3 + s_3 = 60$$

and

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0$$

Notice that we changed the maximization objective to a minimization objective. We also added slack variables to turn the inequalities into equalities.

Ex 2)

Suppose ACME needs to decide which of n different goods to produce. Good i uses a_{ij} units of material j to produce. Each material j has only b_j units available. Good i returns c_i dollars of profit per good produced. If ACME wants to maximize profits, formulate the number of goods to produce as an LP.

General Form Formulation

Let x_i represent the quantity of good x_i to produce

$$\text{maximize } z = \sum_{i=1}^{n} c_i x_i$$

subject to

$$\sum_{j=1}^{m} a_{ij} x_j \leq b_i, \text{ for } i = 1, 2, \ldots, m$$

$$x_i \geq 0 \text{ for } i = 1, 2, \ldots n$$

We can also write this in matrix form. Let

Let $\vec{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$, $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{pmatrix}$, $A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$, and $\vec{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$. Then the problem can be stated as :

$$\text{maximize } f(\vec{x}) = \vec{c} \cdot \vec{x}$$

subject to

$$A\vec{x} \leq \vec{b}$$

$$\vec{x} \geq \vec{0}$$

Finally, we can also write this problem in our index notation. Recall repeated indices in a product indicates a sum, and any generic indices are supposed to carry over the entire range of the index. This can be stated as:

$$\text{maximize } z = c_i x_i$$

subject to

$$a_{ij} x_j \leq b_i$$

$$x_i \geq 0$$

Isn't index notation convenient!

Standard Form Formulation

Now, we can write this in standard form. If we use index notation, we only need to introduce some slack variables s_i.

$$\text{minimize } z = -c_i x_i$$

subject to

$$a_{ij} x_j + s_i = b_i$$

$$x_i, s_i \geq 0$$

We can also state this in matrix form. If we assume all of the variables are the same as the previous matrix form, we only need to introduce the vector

$$\vec{s} = \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_m \end{pmatrix}$$

$$\text{minimize } f(\vec{x}) = -\vec{c} \cdot \vec{x}$$

subject to

$$A\vec{x} + \vec{s} = \vec{b}$$

$$\vec{x}, \vec{s} \geq \vec{0}$$

Or if you insist on using the full summation form:

$$\text{minimize } z = -\sum_{i=1}^{n} c_i x_i$$

subject to

$$s_i + \sum_{j=1}^{m} a_{ij} x_j = b_i, \text{ for } i = 1, 2, \ldots, m$$

$$x_i, s_i \geq 0 \text{ for } i = 1, 2, \ldots n$$